BLOWING UP A SMOOTH CLOSED SUBSCHEME OF A SMOOTH SCHEME STAYS SMOOTH WITH PROJECTIVE EXCEPTIONAL DIVISOR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4974-4988 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 4990-5024, the complete author proof (33 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Blowing up along a closed immersion, conormal sheaves, projective space bundles, and smoothness over a base; the local computation is retained.

Source reference. Original lemma 02OS: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5024–5024. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02NS: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5024–5024. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0805: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5024–5024. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 067U: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5024–5024. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-etale-local-structure: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4861–4878. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma remark-relative-blowup: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4940–4967. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $Z \to X$ be a closed immersion of schemes smooth over S. Let $b : X' \to X$ be the blowing up of Z with exceptional divisor $E \subset X'$. Then X' and E are smooth over S. The morphism $p : E \to Z$ is canonically isomorphic to the projective space bundle*

$$\mathbf{P}(\mathcal{I}/\mathcal{I}^2) \longrightarrow Z$$

where $\mathcal{I} \subset \mathcal{O}_X$ is the ideal sheaf of Z. The relative $\mathcal{O}_E(1)$ coming from the projective space bundle structure is isomorphic to the restriction of $\mathcal{O}_{X'}(-E)$ to E.

Proof. By Divisors, Lemma 067U the immersion $Z \to X$ is a regular immmersion, hence the ideal sheaf $\mathcal{I}$ is of finite type, hence b is a projective morphism with relatively ample invertible sheaf $\mathcal{O}_{X'}(1) = \mathcal{O}_{X'}(-E)$, see Divisors, Lemmas 02OS and 02NS. The canonical map $\mathcal{I} \to b_*\mathcal{O}_{X'}(1)$ gives a closed immersion

$$X' \longrightarrow \mathbf{P}\left(\bigoplus\nolimits_{n\geq 0} \mathrm{Sym}^n_{\mathcal{O}_X}(\mathcal{I})\right)$$

by the very construction of the blowup. The restriction of this morphism to E gives a canonical map

$$E \longrightarrow \mathbf{P}\left(\bigoplus\nolimits_{n\geq 0} \mathrm{Sym}^n_{\mathcal{O}_Z}(\mathcal{I}/\mathcal{I}^2)\right)$$

over Z. Since $\mathcal{I}/\mathcal{I}^2$ is finite locally free if this canonical map is an isomorphism, then the final part of the lemma holds. Having said all of this, now the question is étale local on X. Namely, blowing up commutes with flat base change by Divisors, Lemma 0805 and we can check smoothness after precomposing with a surjective étale morphism. Thus by the étale local structure of a closed immersion of schemes over S given in Lemma lemma-etale-local-structure this reduces us to the case discussed in Remark remark-relative-blowup. □